

Constitutive equations

A summary of constitutive equations useful for modeling, organized by application. Sources are given in parentheses.

H. Heat transfer

H1. Fourier's law - conduction (Incropera, 2003)

$$q_x = -k \cdot \frac{dT}{dx}$$

- q_x is heat flux e.g. W m^{-2}
- k is thermal conductivity e.g. $\text{W m}^{-1} \text{K}^{-1}$
- T is temperature of course, $^{\circ}\text{C}$ or K

← Plane wall, steady state

$$q_x = -k \cdot \Delta T / L$$

- ΔT is temperature difference across wall.
- L is wall thickness.
- $+$ or $-$ is up to you - remember heat flows from hot to cold.

H2. Newton's law of cooling - convection (Incropera, 2003)

$$q = h(T_s - T_{\infty})$$

- q is heat flux from surface
- h is the (convection) heat transfer coefficient

$$q = h \cdot \Delta T$$

- T_s is surface temperature
- T_{∞} is ambient temperature

H3. Overall heat transfer coefficient approach

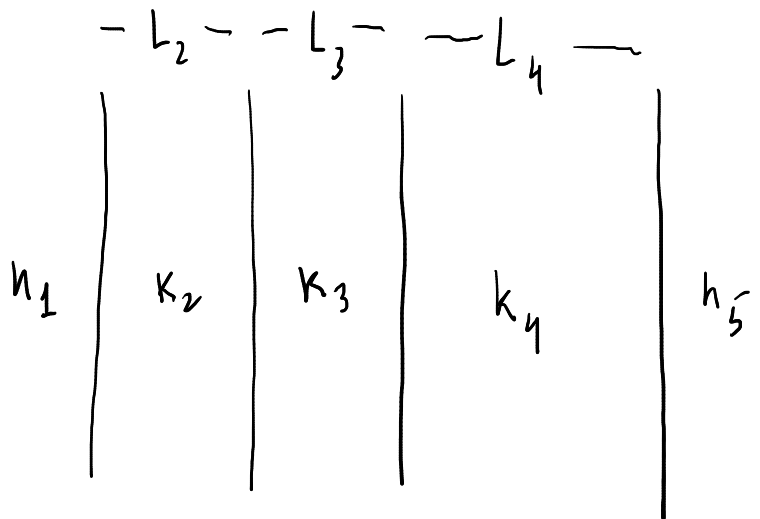
$$q = U \cdot \Delta T$$

- U is overall heat transfer coefficient.

- $q, h, \Delta T$ as above

- + or - up to you but heat always flows from hot to cold!
Use for resistance in series.

$$U = \frac{1}{\frac{1}{h_1} + \frac{L_2}{k_2} + \dots \frac{1}{h_n}}$$



M. Mass transfer

M1. Fick's law - diffusion (Incropera, 2003)

$$J_x = -D \cdot \frac{dc}{dx}$$

J_x = Mass flux . e.g. $\text{g m}^{-2} \text{s}^{-1}$

D is diffusivity e.g. $\text{m}^2 \text{s}^{-1}$

c is concentration e.g. g m^{-3}

$$J_x = -D \cdot \frac{\Delta c}{\Delta x}$$

Plane wall at steady state.

+ D or $-D$ depends on how you define Δc . mass flows from high to low c .

M2. Mass transfer coefficient approach - convection or diffusion (Cussler, 2009)

$$J = k_c \cdot \Delta c$$

k_c is mass transfer coefficient, e.g. m s^{-1}

M3. Interphase mass transfer (through a interface between phases)

$$J = k_L \cdot (c_L - c^*)$$

k_L is an overall mass transfer coefficient in liquid phase units, e.g. m s^{-1}

c_L is bulk liquid phase concentration, e.g. g m^{-3}

c^* is bulk gas phase conc converted to liquid phase

$$J = k_G \cdot (c_G - c^*)$$

k_G is an overall mass transfer coefficient in gas phase units, e.g. m s^{-1} .

c_G is bulk gas phase concentration, e.g. g m^{-3} .

c^* is bulk liquid phase concentration converted to gas phase.

X. Microbial activity

X1. Monod equation - substrate utilization rate (Rittmann and McCarty, 2001)

$$r_{ut} = \frac{\hat{q} \cdot S}{K_{1/2} + S} \cdot X_a$$

r_{ut} is substrate utilization rate, e.g. $g\ m^{-3}\ s^{-1}$.

$\hat{q}$ is maximum (specific) utilization rate, e.g.

$$g\ g_s^{-1}\ s^{-1}.$$

S is substrate concentration, e.g. $g\ m^{-3}$ (usually COD).

$K_{1/2}$ is S concentration that gives 1/2 maximum rate

X_a is active microbial biomass concentration, e.g. $g\ m^{-3}$ (usually COD).